Sensitivity Conjecture

Andreas Lizenberger

University of Siegen

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Abstract

Sensitivity and *block sensitivity* are complexity measures for boolean functions. It has been been an open question whether there is a polynomial relation between these measures. (1) Hao Huang solved a related problem on induced subgraphs of n-dimensional hypercubes and, using a result by (2) Gotsman and Linial, (3) Nisan and Szegedy, proved the following relation for all boolean functions f:

$$\exists \alpha > 0 \colon \forall f \colon bs(f) \leq s(f)^{\alpha}$$

Table of Contents

Boolean Functions

Definition

A boolean function is a function $f\colon \{\text{false}, \text{true}\}^n \to \{\text{false}, \text{true}\}$.

We will use *true* $\equiv 1$ and
depending on the context *false* $\equiv 0$ or alternatively *false* $\equiv -1$.

Example

We can express logical *AND* as a boolean function:

$$AND_3(x) = \begin{cases} 1, & \text{if } x = (1,1,1) \\ 0, & \text{otherwise} \end{cases}$$

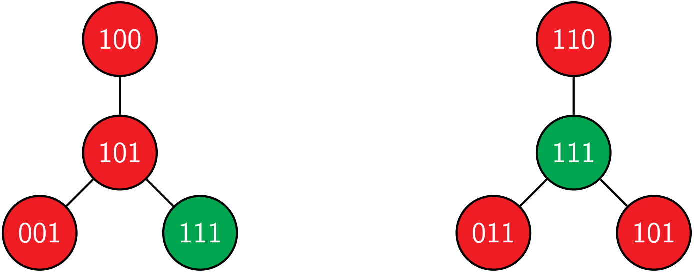
We will write tuples in their bit string representation when possible:

$$(1,1,1) \equiv 111$$

Sensitivity

Local sensitivity for the boolean function AND_3 and inputs $101, 111$:

- $AND_3(101) = 0 \Rightarrow$ colored red, $AND_3(111) = 1 \Rightarrow$ colored green



We must count the number of neighbors with a different value:

- Sensitivity $s(AND_3, 101) = 0 + 1 + 0 = 1$
- Sensitivity $s(AND_3, 111) = 1 + 1 + 1 = 3$

Sensitivity

Sensitivity for the boolean function AND_3:

$$s(AND_3, 000) = 0 \quad s(AND_3, 001) = 0 \quad s(AND_3, 010) = 0 \quad s(AND_3, 011) = 1$$

$$s(AND_3, 100) = 0 \quad s(AND_3, 101) = 1 \quad s(AND_3, 110) = 1 \quad s(AND_3, 111) = 3$$

We are interested in the biggest sensitivity over all inputs x:

$$s(AND_3) = \max_{x \in \{0,1\}^3} s(AND_3, x) = s(AND_3, 111) = 3$$

Sensitivity

Definition

For a boolean function $f: \{0,1\}^n \to \{0,1\}$ the sensitivity $s(f)$ is:

$$s(f) = \max_{x \in \{0,1\}^n} s(f, x)$$

In this definition we defined the neighbors U for an input x implicitly. For block sensitivity we must define neighbors explicitly.

Block Sensitivity

Let $x = x_n x_{n-1} \dots x_1$ be an input of length n and $1 \leq k \leq n$:

$$\mathit{flip}_n(x, k) := x_n x_{n-1} \dots (1 - x_k) \dots x_1$$

Let $K = \{k_1, k_2, \dots, k_m\} \subseteq \mathcal{I} = \{1, 2, \dots, n\}$:

$$\mathit{flip}(x, K) := \mathit{flip}_n(\dots \mathit{flip}_n(\mathit{flip}_n(x, k_1), k_2) \dots, k_m)$$

We have already implicitly used the function *flip*. Now we formally generalized it for index sets K. Example:

$$\mathit{flip}(100, \{1, 2\}) = 111$$

Block Sensitivity

Let $\mathcal{K} = \{K_1, K_2, \ldots, K_m\}$ be a partition of $\mathcal{I} = \{1, 2, \ldots, n\}$. A partition has the following properties:

- $K_i \neq \emptyset, \quad 1 \leq i \leq m$
- $\bigcup_{i=1}^{m} K_i = \mathcal{I}$
- $K_i \cap K_j = \emptyset, \quad \forall i \neq j$

We have already implicitly used the partition $\{\{1\}, \{2\}, \ldots, \{n\}\}$.

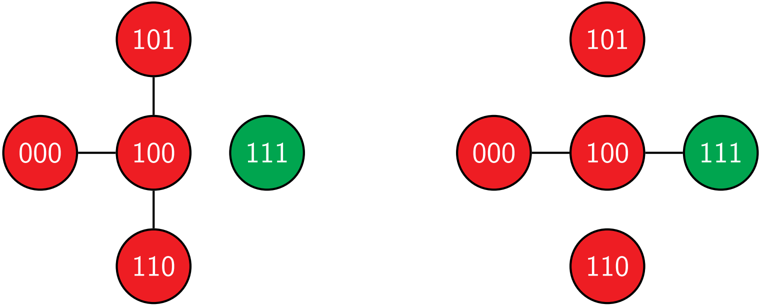
We define the set $U(x, \mathcal{K})$ for input $x \in \{0,1\}^n$:

$$U(x, \mathcal{K}) = \{flip(x, K) \mid K \in \mathcal{K}\}$$

and call it the neighbors of x for partition $\mathcal{K}$.

Block Sensitivity

Let $\mathcal{K}_1 = \{\{1\},\{2\},\{3\}\}$, $\mathcal{K}_2 = \{\{1,2\},\{3\}\}$ and $x = 100$.



We find for AND_3 that the sensitivity for the neighbors $U(x,\mathcal{K}_1)$ is bigger than the sensitivity for the neighbors $U(x,\mathcal{K}_2)$.

Block Sensitivity

Definition (verbose)

Let $\mathcal{K} = \{\{1\}, \{2\}, \ldots, \{n\}\}$.
For a boolean function $f \colon \{0,1\}^n \to \{0,1\}$ the sensitivity $s(f)$ is:

$$s(f) = \max_{x \in \{0,1\}^n} s(f, x, U(x, \mathcal{K}))$$

Definition

Let $\mathcal{I} = \{1, 2, \ldots, n\}$.
For a boolean function $f \colon \{0,1\}^n \to \{0,1\}$ the block sensitivity $bs(f)$ is:

$$bs(f) = \max_{\substack{\mathcal{K} \text{ is a} \\ \text{partition of } \mathcal{I}}} \max_{x \in \{0,1\}^n} s(f, x, U(x, \mathcal{K}))$$

A direct consequence of the previous definitions is:

$$s(f) \leq bs(f)$$

Our goal is to prove the sensitivity conjecture.

Theorem

The sensitivity conjecture that states that for all boolean functions f:

$$\exists \alpha > 0\colon \forall f\colon bs(f) \leq s(f)^{\alpha}$$

Let us outline the results that (2) Gotsman and Linial and (3) Nisan and Szegedy have established.

Nisan and Szegedy

Theorem

For every boolean function f we can find a unique multi-linear polynomial $\hat{f}$ such that $\forall x \in \{-1,1\}^n$: $f(x) = \hat{f}(x)$.

Definition

The degree $deg(f) = deg(\hat{f})$ of a boolean function f is the maximum number of variables occurring in any monomial in $\hat{f}$.

Example

The multi-linear polynomial $\widehat{AND_2}(x_2, x_1) = \frac{1}{2} \cdot x_2 \cdot x_1 + \frac{1}{2} \cdot x_2 + \frac{1}{2} \cdot x_1 + 0$ represents the boolean function AND_2. The degree of AND_2 is 2.

Nisan and Szegedy

Nisan and Szegedy found the following relation between the degree and the block sensitivity.

Theorem

For all boolean functions f it holds:

$$\sqrt{\frac{1}{2}bs(f)} \leq deg(f)$$

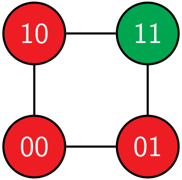
We only need to find the relation between $deg(f)$ and $s(f)$.

Gotsman and Linial

Definition

Let $Q_n = (V, E)$ be a graph that consists of the vertices $V = \{-1, 1\}^n$ and edges $E = \{(v_1, v_2) \in V^2 \mid \exists k\colon \mathit{flip}_n(v_1, k) = v_2\}$.
We call Q_n the n-dimensional hypercube for $n > 3$ and cube for $n = 3$.

For $n = 2$ we get:

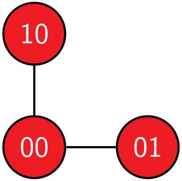


Gotsman and Linial

Definition

The induced subgraph $\hat{G}$ of a graph $G = (V, E)$ for a vertex set $W \subseteq V$ is the graph $\hat{G} = (W, \hat{E})$, with $\hat{E} = \{(v_1, v_2) \in W^2 \mid (v_1, v_2) \in E\}$.

If we remove the vertex 11 we must remove the corresponding edges:



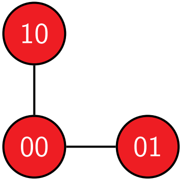
Gotsman and Linial

Definition

Let $G = (V, E)$ be a graph. The degree $deg_G(v)$ of a vertex $v \in V$ is the number of neighbors of v in E. The degree $\Delta(G)$ of the graph G is:

$$\Delta(G) = \max_{v \in V} deg_G(v)$$

In our example: $\Delta(\hat{G}) = 2 = deg_{\hat{G}}(00)$, $deg_{\hat{G}}(01) = 1 = deg_{\hat{G}}(10)$.



Gotsman and Linial

Gotsman and Linial found the following relation between the sensitivity and the degree.

Theorem

Let Q_n be the n-dimensional hypercube. Let $h\colon \mathbb{N} \to \mathbb{R}$ be a function. The following statements are equivalent:

- for all boolean functions f it holds: $h(deg(f)) < s(f)$
- for all induced subgraphs $\hat{Q}_n = (W, \hat{E})$ of $Q_n = (V, E)$ with $|W| > 2^{n-1}$ it holds: $h(n) \leq \Delta(\hat{Q}_n)$

This means we can prove the first statement by proving the second statement instead. If we select $h(n) = \sqrt{n}$ then we get the following result:

$$\sqrt{\frac{1}{2} bs(f)} \leq deg(f) < s(f)^2$$

Huang

Hao Huang proved the second statement for $h(n) = \sqrt{n}$ and therefore the Sensitivity Conjecture.

Theorem

Let Q_n be the n-dimensional hypercube.
For all induced subgraphs $\hat{Q}_n = (W, \hat{E})$ of $Q_n = (V, E)$ with $|W| = 2^{n-1} + 1$ it holds: $\sqrt{n} \leq \Delta(\hat{Q}_n)$

Remarks:
Without loss of generality we can assume that the induced subgraphs have exactly $2^{n-1} + 1$ many vertices.
We can also simplify the proof by constructing a vertex v_α with:

$$\sqrt{n} \leq deg_{\hat{Q}_n}(v_\alpha)$$

Huang

Theorem

Let Q_n be the n-dimensional hypercube.
For all induced subgraphs $\hat{Q}_n = (W, \hat{E})$ of $Q_n = (V, E)$ with $|W| = 2^{n-1} + 1$ it holds: $\sqrt{n} \leq \Delta(\hat{Q}_n)$

Proof Outline:

- There is an adjacency matrix A_n with eigenvalues $\pm\sqrt{n}$.
- There is a matrix B_n whose columns are eigenvectors of A_n to the eigenvalue $\sqrt{n}$.
- There is a vector $z \neq 0$ such that:
  - z is a linear combination of the column vectors of B_n
  - vertex $v \notin W \Rightarrow$ the corresponding entry $(z)_v = 0$
- There is an index α: $max\{|z_1|, |z_2|, \cdots, |z_{2^n}|\} = |(z)_\alpha| > 0$.
- It holds: $|\sqrt{n}\,(z)_\alpha| = |(A_n z)_\alpha| \leq deg_{\hat{Q}_n}(v_\alpha)\,|(z)_\alpha|$, with $v_\alpha \in W$

Huang

Proof:
There is an adjacency matrix A_n with eigenvalues $\pm\sqrt{n}$.
We inductively define:

$$A_n = \begin{pmatrix} A_{n-1} & Id_{2^{n-1}} \\ Id_{2^{n-1}} & -A_{n-1} \end{pmatrix}$$

where Id is the identity and $A_0 = (0)$. For $n = 2$ we get:

$$A_2 = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & -1 \\ 0 & 1 & -1 & 0 \end{pmatrix}$$

Using induction (ind) we can show:

$$A_n^2 = n\, Id_{2^n}$$

We find:

$$\begin{aligned} A_n^2 &= \begin{pmatrix} A_{n-1} & Id_{2^{n-1}} \\ Id_{2^{n-1}} & -A_{n-1} \end{pmatrix} \begin{pmatrix} A_{n-1} & Id_{2^{n-1}} \\ Id_{2^{n-1}} & -A_{n-1} \end{pmatrix} \\ &= \begin{pmatrix} A_{n-1}^2 + Id_{2^{n-1}} & A_{n-1} - A_{n-1} \\ A_{n-1} - A_{n-1} & Id_{2^{n-1}} + A_{n-1}^2 \end{pmatrix} \overset{ind}{=} \begin{pmatrix} n\, Id_{2^{n-1}} & 0 \\ 0 & n\, Id_{2^{n-1}} \end{pmatrix} \end{aligned}$$

Assume there exist eigenvectors $v \neq 0$ to eigenvalues λ:

$$n\, Id_{2^n} v = A_n^2\, v = \lambda^2 v \;\Rightarrow\; \lambda_{1,2} = \pm\sqrt{n}$$

Huang

There is a matrix B_n whose columns are eigenvectors of A_n to the eigenvalue $\sqrt{n}$.
We inductively define:

$$B_n = \begin{pmatrix} A_{n-1} + \sqrt{n}\, Id_{2^{n-1}} \\ Id_{2^{n-1}} \end{pmatrix}$$

We find:

$$\begin{aligned} A_n B_n &= \begin{pmatrix} A_{n-1} & Id_{2^{n-1}} \\ Id_{2^{n-1}} & -A_{n-1} \end{pmatrix} \begin{pmatrix} A_{n-1} + \sqrt{n}\, Id_{2^{n-1}} \\ Id_{2^{n-1}} \end{pmatrix} \\ &= \begin{pmatrix} A_{n-1}^2 + \sqrt{n}\, A_{n-1} + Id_{2^{n-1}} \\ A_{n-1} + \sqrt{n}\, Id_{2^{n-1}} - A_{n-1} \end{pmatrix} = \begin{pmatrix} n\, Id_{2^{n-1}} + \sqrt{n}\, A_{n-1} \\ \sqrt{n}\, Id_{2^{n-1}} \end{pmatrix} \\ &= \begin{pmatrix} \sqrt{n}(A_{n-1} + \sqrt{n}\, Id_{2^{n-1}}) \\ \sqrt{n}\, Id_{2^{n-1}} \end{pmatrix} = \sqrt{n}\, B_n \end{aligned}$$

Huang

There is a vector $z \neq 0$ such that:

- z is a linear combination of the column vectors of B_n
- vertex $v \notin W \Rightarrow$ the corresponding entry $(z)_v = 0$

We can use vertices to index rows:

$$
\begin{gathered}
(-1,-1,-1) \equiv (0,0,0) \equiv 1 \\
(-1,-1,1) \equiv (0,0,1) \equiv 2 \\
\vdots \\
(1,1,1) \equiv (1,1,1) \equiv 8
\end{gathered}
$$

Huang

Let B^* be the matrix with rows of B_n whose indices are not in the induced subgraph $\hat{Q}_n$. We can find an $x \in \mathbb{R}^{2^{n-1}} \neq 0$ such that: $B^* x = 0$, because of the Rank-Nullity Theorem:

Theorem

Let $A\colon M \to N$ be a linear mapping between two vector spaces of finite dimension. The Rank-Nullity Theorem says:

$$rank(A) + nullity(A) = dim(M)$$

B^* has $rank(B^*) \leq 2^{n-1} - 1 \Rightarrow nullity(B^*) \geq 2^{n-1} - 2^{n-1} + 1 \geq 1$.

Define z through the linear combination:

$$z = B_n x \neq 0$$

Huang

There is an index α: $max\{|z_1|, |z_2|, \cdots, |z_{2^n}|\} = |(z)_\alpha| > 0$.
We find:

$$|(A_n z)_\alpha| = |(A_n)_\alpha\, z| = \left|\sum_{i=0}^{2^n} (A_n)_{\alpha,i}\,(z)_i\right| \leq \sum_{i=0}^{2^n} |(A_n)_{\alpha,i}|\;|(z)_i|$$

By definition $(z)_i = 0$ if i is not a vertex in the induced subgraph $\hat{Q}_n$:

$$\sum_{i=0}^{2^n} |(A_n)_{\alpha,i}|\;|(z)_i| = \sum_{\substack{i=0\\ v_i \in W}}^{2^n} |(A_n)_{\alpha,i}|\;|(z)_i| + \underbrace{\sum_{\substack{i=0\\ v_i \notin W}}^{2^n} |(A_n)_{\alpha,i}|\;|(z)_i|}_{=0}$$

Huang

Using the definition $|(z)_i| \leq |(z)_\alpha|$:

$$\sum_{\substack{i=0 \\ v_i \in W}}^{2^n} |(A_n)_{\alpha,i}|\;|(z)_i| \leq |(z)_\alpha| \sum_{\substack{i=0 \\ v_i \in W}}^{2^n} |(A_n)_{\alpha,i}|$$
$$= |(z)_\alpha|\, deg_{\hat{Q}_n}(v_\alpha)$$

This gives us:

$$|(A_n z)_\alpha| \leq |(z)_\alpha|\, deg_{\hat{Q}_n}(v_\alpha)$$

Furthermore, it holds:

$$\sqrt{n}\,|(z)_\alpha| = |(\sqrt{n}\,z)_\alpha| = |(A_n z)_\alpha|$$

because z is a linear combination of eigenvectors.

Huang

This proves:

$$
\begin{aligned}
& \sqrt{n}\,|(z)_\alpha| \leq |(z)_\alpha|\, deg_{\hat{Q}_n}(v_\alpha) \\
\Rightarrow \quad & \sqrt{n} \leq deg_{\hat{Q}_n}(v_\alpha) \\
& \sqrt{n} \leq \Delta(\hat{Q}_n))
\end{aligned}
$$

$\square$

Ending

Thank you for your attention.

Are there any questions?

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